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Adaptive Context-Aware Malicious Flooding Detection and Mitigation in Vehicular Communication Networks

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1. Title

Adaptive Context-Aware Malicious Flooding Detection and Mitigation in Vehicular Communication Networks

2. Abstract

Detection of malicious flooding is hard to perform with vehicular communication networks since normal congestion may result in communication surges that look like malicious traffic. The base paper addresses the issue with a contextual anomaly detection approach and spatiotemporal clustering, localized contextual baseline, session formation, and receiver-window formation. Experiments in the base paper show that contextual representation helps detect malicious flooding using machine learning better than traffic representation without context. However, the base paper relies on controlled and simulated dataset and does not provide an adaptive solution to the problem. Its contextual approach additionally imposes extra computational needs for the roadside infrastructure. Changing traffic conditions may make a normal communication profile change. In order to overcome those shortcomings, this work suggests an Adaptive Context-Aware Malicious Flooding Detection and Mitigation (ACAFDM) framework. Proposed framework uses spatiotemporal and congestion-aware representations and a rolling local baseline that updates with recent validated benign traffic. Contextual deviation features are created using transmission rate, payload size, concurrency, mobility, and receiver-window behaviors. Then a lightweight Gradient Boosting model is used to estimate the probability of malicious flooding. Unlike the base paper where an immediate binary blocking decision was made, proposed solution involves a confidence-aware decision layer that distinguishes between benign traffic, suspicious traffic, and high-confidence malicious traffic. A selective mitigation module at roadside unit performs the rate-limiting and deprioritization of high-confidence malicious traffic while maintaining the safe communication. Proposed framework will be evaluated using different highway and metropolitan scenarios, different flooding intensities and changing traffic using metrics of detection accuracy, precision, recall, F1-score, false-positive rate, detection delay, and mitigation overhead. Expected outcome is a lightweight and adaptive extension of congestion-context-aware malicious flooding

detection that becomes more robust to changing traffic and takes the next step to intrusion detection system development.

3. Keywords

Vehicular communication networks, Malicious flooding, Congestion-aware detection, Spatiotemporal analysis, Machine learning, Adaptive baseline

4. Introduction

Vehicular Communication Networks (VCNs) are becoming a critical part of communication infrastructure for Intelligent Transportation Systems since they facilitate exchange of information between vehicles and other vehicles, roadside infrastructure, and network service providers. Such communication is essential for applications including awareness, traffic control, road condition information, collision avoidance, and others. Contrary to traditional fixed network communications, vehicular networks operate in a dynamic environment where the density, positions, speeds, connectivity and communication relationships between vehicles are changing all the time. Hence, security measures for VCNs should provide protection from attacks while providing the necessary reliability and low latency for vehicular applications.

Among the important availability threats in VCNs are malicious flooding attacks. During a flooding attack, one or several compromised vehicles flood the network with a lot of traffic in order to consume the available bandwidth, cause the congestion, occupy the receiver or forwarding resources and interrupt communication. This attack is usually revealed in abnormally high rates of packets, bytes, communication concurrency or repeated traffic directed towards receiver. However, the high communication activity itself is not the sign of malicious behavior. A metropolitan intersection during the rush hour, a traffic jam, or densely packed vehicles can naturally cause an increased level of communication activity. As a result, a detector that relies on a global rate threshold will consider the natural congestion as an attack, while a sophisticated or low-rate attack will remain under this threshold.

This dilemma of distinguishing the congestion from an attack is the main issue considered in the selected base paper – "Congestion-context-aware spatiotemporal detection of malicious flooding in vehicular communication networks." The base paper considers flooding attacks as the contextual anomalies rather than a simple rate-threshold problem. The main idea of the paper is that the behavior of communication should be interpreted depending on the context of space, time, mobility and congestion at which the communication occurred. It is very important to do so since the same transmission rate can be normal in some circumstances and suspicious in another. High rate of transmission in a congested metropolitan area is not a surprise, but in a relatively non-congested local area the same rate becomes suspicious.

For this purpose, the base paper introduces a congestion context-aware framework based on several complementary representation forms. The spatiotemporal clustering provides the local

context of communication. The localized contextual baselines define the expected communication activity in the context. The session formation combines the communication between transmitter and receiver entities. Receiver window formation provides the receiver-based representation of the communication activity inside the spatiotemporal cluster. Thus, the detector is able not only to analyze the amount of the traffic caused by the communication flow, but also to check how the traffic relates to the surrounding traffic situation.

The associated dataset provides support to this approach by including the spatial, temporal, mobility, communication, contextual, and identification information. The spatial information includes the source and destination positions and the distance between them. The temporal information includes the beginning and ending time of the transmission, flow duration and receiver window span. The mobility information includes the vehicle speed and the distance-related characteristics of the movement. The communication information includes the number of transmitted and received bytes and packets, packet loss, delay, jitter, throughput, delivery ratio, hop count, transmission rate, payload size and concurrency. The contextual information includes the background transmission rate, background payload size and background concurrency.

The base paper tests the proposed method on highways and metropolitan mobility contexts and varies the intensity of the malicious attacks. The authors demonstrate that the representation of context helps to improve machine-learning-based detection of malicious flooding in the considered environments. Among the applied classifiers, contextual Gradient Boosting shows the best performance. The impact of contextual information is further analysed using an ablation study, where the contextual features are excluded. A two-tailed McNemar test is applied to verify whether the performance difference is statistically significant. According to the presented results, the authors prove the main hypothesis concerning the importance of congestion and mobility context for detecting malicious traffic.

Nevertheless, even though the base paper is solidly designed, there are some limitations that can be observed in terms of its scope and experimental setup. First, the use of simulated data is a limitation of the evaluation. Simulated data allows the authors to conduct reproducible experiments and work with clearly defined attack scenarios, but actual vehicular networks could include such issues as noisy data, missed observations, unexpected behaviour of vehicles, variations in traffic demands, or even attacks that do not match the existing attack scenarios. Thus, good performance on the provided scenarios is not enough to claim robustness of the solution in terms of all possible traffic conditions.

Secondly, the main goal of the base paper is detection and classification, whereas an integrated detection-and-mitigation process is needed. It is true that finding the source of the attack is an important step. Still, a real-world roadside security mechanism must choose what to do after the attack is detected. Blocking all messages of the detected suspect is not a good idea since any detector can produce false positives during legitimate traffic surges. Also, vehicular networks include messages with various priorities and it is unacceptable to block all traffic. Therefore, a

reasonable solution must have a selective response algorithm that reduces the amount of malicious traffic while preserving important communications.

Thirdly, the problem of the changing character of vehicular traffic is another limitation. Normal communication is not stationary. The number of vehicles may grow during rush hours, decrease after clearing a traffic jam or even vary depending on events happening on the road or other reasons. A contextual baseline representing normal communication in a certain traffic period becomes less representative later. If it is not updated, then a detector will either produce false positive detections or will be unable to detect an attack when it occurs against the background of a new traffic pattern. Hence, an adaptive mechanism is needed to constantly update legitimate changes in communication traffic.

Fourthly, it can be stated that there are some processing needs of contextual representation. Spatiotemporal clustering, construction of a contextual baseline, session formation, and receiver window formation are important operations providing useful information. They should be done continuously in case if the detector is running on a roadside unit. Such a mechanism may have to work with communication of multiple vehicles at once. Thus, it is necessary to maintain low detection latency while keeping the advantage of using contextual information.

Finally, there is the issue of generalization beyond the evaluated environments. While the highway and metropolitan environments and different flooding intensities introduced by the base paper serve as a valuable testbed for evaluating the proposed work, further testing should be done in other traffic conditions and with attacks that have never been observed before. The model may perform well with familiar distributions but fail to cope with new distributions when the underlying traffic profile changes. Thus, it is reasonable to test the robustness of the work not only for classification but also for adaptability.

In order to solve the listed problems, an Adaptive Context-Aware Malicious Flooding Detection and Mitigation (ACAFDM) framework is proposed. The proposed approach will extend the main contribution of the base paper rather than replace it. Namely, spatiotemporal clustering, contextual representation, session formation, receiver-window formation, and related communication and mobility features are taken as the basis of the proposed approach. In addition, three more elements are introduced – adaptive local baseline, confidence-aware detection stage, and selective mitigation stage.

The purpose of adaptive local baseline is to deal with changes in the behavior of vehicular traffic. For every local spatiotemporal context, there will be maintained recent estimations of normal communication behavior. For example, statistics on the rate of transmissions, size of payload, concurrency, receiver activity, and other contextual characteristics can be recalculated based on recent validated benign communications. Then, the current communication behavior can be compared with the recent baseline behavior instead of fixed historical profile. As a result, contextual-deviation features, indicating the deviation from the local state, will be received.

In order to implement the detection stage, a lightweight Gradient Boosting classifier will be used. The base paper shows that Gradient Boosting works effectively in the context of the problem under consideration. The classifier will get input from communication, spatial, temporal, mobility, contextual, and contextual-deviation features. The classifier will return not a binary result but attack confidence value. Thus, the decision-making algorithm will be able to distinguish between normal traffic, uncertain/suspicious traffic, and malicious traffic.

As for the confidence-aware decision-making, it should minimize the unnecessary mitigation of the traffic which is actually legitimate. Communication with the low attack confidence can be sent further using the usual forwarding procedure. The intermediate level of attack confidence can put the traffic in the suspicious state where it should be analyzed using further observations. Only persistent high-confidence malicious traffic should be forwarded to the mitigation stage.

The mitigation phase is another major extension of the base paper. After a flooding attack has been detected with enough confidence, the roadside unit is able to use selective rate limiting or traffic deprioritization for the suspicious traffic. The idea is to decrease the resource usage of malicious flooding attacks without denying legitimate communication attempts from the source node. The safety-critical traffic is not necessarily affected, whereas the high-volume suspicious traffic is deprioritized or limited in its rate. Thus, the proposed system integrates attack detection with a concrete protection method.

5. Literature Survey

This section reviews twenty peer-reviewed and preprint research works closely related to the base paper, spanning VANET and vehicular-network security surveys, DDoS and flooding-attack detection, SDN-based vehicular defense mechanisms, and general machine-learning and deep-learning intrusion detection techniques applicable to vehicular contexts. The papers are presented first in tabular form for rapid comparison, and subsequently as individual academic paragraphs summarizing the methodology, contribution, and limitations of each work in relation to the base paper's congestion-context-aware approach.

5.1 Literature Survey (Table Format)

S.No	Paper Title	Authors	Year	Methodology	Key Findings	Advantages	Limitations
1	Detection of DDoS Attacks in SDN-Based VANET Using Optimized TabNet	Setitra, M. A. and Fan, M.	2024	Combines a Software-Defined Networking (SDN) control plane with the TabNet deep tabular-learning architecture, tuned through a metaheuristic optimizer, to classify flow-level traffic features collected at SDN controllers serving vehicular nodes.	The optimized TabNet classifier attains high detection accuracy while reducing controller overhead compared with conventional CNN and LSTM baselines, and it adapts quickly to changing traffic patterns typical of vehicular flows.	Centralized SDN visibility simplifies feature collection and enables rapid, network-wide policy updates.	Reliance on a centralized SDN controller creates a single point of failure and does not explicitly separate benign congestion bursts from attack traffic.
2	Real-Time DDoS Flooding Attack Detection in Intelligent Transportation Systems	Karthikeyan, H. and Usha, G.	2022	Proposes a real-time entropy-based statistical detection scheme that monitors packet arrival rates and header entropy at roadside infrastructure to flag abrupt traffic surges as flooding events.	The scheme detects high-rate flooding attacks with low latency, making it suitable for time-critical Intelligent Transportation System (ITS) applications such as collision warning.	Lightweight, low-latency detection appropriate for resource-constrained roadside units.	Entropy-based thresholds are sensitive to natural fluctuations in traffic density and do not incorporate spatial or mobility context, leading to elevated false positives under congestion.
3	Intrusion Prevention System for DDoS Attack on VANET With reCAPTCHA Controller Using Information Based Metrics	Poongodi, M., Vijayakumar, V., Al-Turjman, F., Hamdi, M., and Ma, M.	2019	Introduces an intrusion prevention system that combines information-theoretic traffic metrics with a reCAPTCHA-style challenge-response controller to filter automated flooding traffic before it reaches RSUs.	The reCAPTCHA controller substantially reduces the volume of malicious traffic that reaches the network core while information-based metrics improve discrimination between legitimate and malicious flows.	Proactive prevention rather than purely reactive detection reduces downstream congestion.	Challenge-response mechanisms introduce latency unsuitable for safety-critical low-latency VANET messages and assume human-operated end devices, which does not hold for autonomous vehicle-generated traffic.
4	Dense Capsule Stacked Auto-Encoder Model Based DDoS Attack Detection and Hybrid Optimal Bandwidth Allocation With Routing in VANET Environment	Tanati, M. K. and Ponnusamy, M.	2025	Employs a dense capsule network stacked with an autoencoder to learn hierarchical spatial-relationship features of vehicular traffic for DDoS detection, coupled with a hybrid metaheuristic bandwidth-allocation and routing scheme.	The capsule-autoencoder hybrid captures part-whole relationships among traffic features, improving detection accuracy over conventional autoencoder or CNN baselines, while joint bandwidth optimization reduces post-detection congestion.	Reconstruction-error-based anomaly scoring combined with capsule routing improves sensitivity to subtle, low-rate flooding patterns.	High model complexity and training cost may be impractical for real-time deployment on resource-limited RSUs; the framework does not explicitly separate congestion-induced reconstruction error from attack-induced error.
5	PETRAK: A Solution Against DDoS Attacks in Vehicular Networks	Verma, A., Saha, R., Kumar, G., and Conti, M.	2024	Proposes a lightweight cryptographic-cum-statistical filtering scheme (PETRAK) that verifies message authenticity while monitoring per-vehicle transmission rates to isolate flooding sources in real time.	PETRAK achieves a favorable balance between detection accuracy and communication overhead, isolating flooding vehicles within a few message rounds without heavy cryptographic cost.	Low computational and bandwidth overhead makes the scheme deployable on resource-constrained on-board units.	Per-vehicle rate thresholds are static and environment-agnostic, so they do not adapt to varying road congestion levels across urban, highway, or dense intersection scenarios.
6	RBF-SVM Kernel-Based Model for Detecting DDoS Attacks in SDN Integrated Vehicular Network	Anyanwu, G. O., Nwakanma, C. I., Lee, J. M., and Kim, D. S.	2023	Applies a Radial Basis Function Support Vector Machine (RBF-SVM) with hyperparameter grid search to classify SDN-collected flow statistics from vehicular networks into benign or DDoS traffic classes.	The tuned RBF-SVM model achieves high classification accuracy on benchmark SDN-vehicular datasets and generalizes well across several attack intensities.	Kernel-based SVM is robust to moderate class imbalance and requires comparatively little training data relative to deep models.	SVM training complexity scales poorly with very large flow datasets, and the model treats each flow independently without modeling spatial correlation among neighboring vehicles.
7	Machine Learning-Enhanced DDoS Attack Detection and Mitigation in VANET Infrastructure	Gayathri, S. et al.	2025	Proposes a Hybrid Deep Learning with Federated Learning (HDL-FL) framework that uses convolutional neural networks to capture spatial and temporal traffic patterns while training is distributed across RSUs and vehicles via federated learning.	HDL-FL improves detection accuracy substantially over centralized baselines, reduces false positives, and increases attack-detection rate while preserving data privacy through federated aggregation.	Federated training preserves data privacy and reduces the communication burden of transmitting raw traffic to a central server.	Federated aggregation introduces convergence delays under non-independent and identically distributed (non-IID) traffic across RSUs and does not explicitly model road-level congestion context.

S.No	Paper Title	Authors	Year	Methodology	Key Findings	Advantages	Limitations
8	Lightweight Statistical Approach towards TCP SYN Flood DDoS Attack Detection and Mitigation in SDN Environment	Batool, S. et al.	2022	Uses lightweight statistical measures (SYN-to-ACK ratio and half-open connection counts) computed at the SDN controller to detect TCP SYN flood attacks and trigger flow-table mitigation rules.	The statistical detector identifies SYN flood onset within a small number of sampling intervals and mitigation reduces controller and switch resource exhaustion significantly.	Very low computational cost enables deployment on constrained SDN controllers without specialized hardware.	Method is specific to TCP-based flooding and does not generalize to broadcast or beacon-flooding patterns typical of vehicular safety-message channels.
9	Detection and Mitigation of SYN Flooding Attacks through SYN/ACK Packets and Black/White Lists	Yang, C. H., Wu, J. P., Lee, F. Y., Lin, T. Y., and Tsai, M. H.	2023	Combines real-time SYN/ACK packet-pair analysis with dynamically maintained black and white lists of source addresses to detect and filter SYN flooding traffic at network edge devices.	The combined statistical-and-listing approach achieves high precision in distinguishing flooding sources from legitimate clients and adapts more quickly than static blacklist-only methods.	Dynamic list maintenance reduces false positives over time as legitimate but previously flagged sources are readmitted.	Blacklisting approaches assume relatively stable source identities, an assumption that is often violated in VANETs where vehicle identifiers and network positions change rapidly due to mobility.
10	A review of security attacks and intrusion detection in the vehicular networks	Nandy, T., Noor, R. M., Kolandaisamy, R., Idris, M. Y. I., and Bhattacharyya, S.	2024	Provides a comprehensive survey and taxonomy of security attacks against vehicular networks, along with a structured review of intrusion detection system (IDS) design paradigms, spanning signature-based, anomaly-based, and hybrid approaches.	The survey identifies flooding and DDoS attacks as among the most disruptive threats to VANET availability and finds that most existing IDS approaches rely on flat, rate-based indicators rather than contextual or environmental features.	Offers a broad, structured taxonomy that helps position individual detection techniques within the wider VANET security landscape.	As a survey, it does not propose or validate a new detection technique; it identifies but does not resolve the gap between rate-based and context-aware detection.
11	Enhancing intrusion detection: a hybrid machine and deep learning approach	Sajid, M. et al.	2024	Combines classical machine-learning classifiers with deep neural feature extractors in a two-stage hybrid pipeline, applying the approach to network intrusion detection benchmark datasets including flooding-style attacks.	The hybrid pipeline outperforms either the classical or the deep component used in isolation, indicating complementary strengths between hand-crafted statistical features and learned deep representations.	Two-stage hybridization balances interpretability of classical features with the representational power of deep models.	Evaluated primarily on generic cloud/network intrusion datasets rather than vehicular-specific traffic, limiting direct transferability to VANET congestion dynamics.
12	An authentication approach in SDN-VANET architecture with Rider-Sea Lion optimized neural network for intrusion detection	Pulligilla, M. K. and Vanmathi, C.	2023	Integrates a bio-inspired Rider-Sea Lion optimization algorithm with a neural-network classifier to jointly optimize authentication and intrusion detection in an SDN-VANET architecture.	The optimized neural network improves detection accuracy over unoptimized baselines and reduces authentication overhead through joint parameter tuning.	Metaheuristic optimization improves convergence speed and avoids poor local minima in neural network training.	The bio-inspired optimizer adds tuning complexity and computational cost during training, and the detection features remain largely traffic-rate oriented rather than congestion-context oriented.
13	An Adaptive Real-Time Malicious Node Detection Framework Using Machine Learning in Vehicular Ad-Hoc Networks (VANETs)	Authors reported in PMC10007041	2023	Proposes an adaptive real-time framework that continuously updates its machine-learning detection model as traffic conditions evolve, targeting malicious nodes responsible for DDoS-style flooding in VANETs.	The adaptive updating mechanism sustains detection accuracy across changing traffic conditions better than statically trained models evaluated once and never retrained.	Continual model adaptation makes the framework more resilient to concept drift caused by evolving traffic patterns.	Frequent model updates increase computational and communication overhead, and the adaptation criteria are driven by classification confidence rather than explicit congestion metrics.
14	A Spatiotemporal-Oriented Deep Ensemble Learning Model to Defend Link Flooding Attacks in IoT Network	Chen, Y. H., Lai, Y. C., Jan, P. T., and Tsai, T. Y.	2021	Combines Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) components in a deep ensemble to jointly capture temporal traffic dynamics and spatial link-topology patterns for link flooding attack (LFA) detection.	The spatiotemporal ensemble improves detection of the changing, adaptive attack characteristics of link flooding attacks compared with models that consider only temporal or only spatial features in isolation.	Explicit joint modelling of spatial (topology) and temporal (traffic-rate) dimensions improves robustness against attackers who deliberately vary their flooding pattern over time and location.	Designed for general IoT/backbone network topologies rather than the highly dynamic, mobility-driven topology of vehicular networks, where the notion of a 'link' is transient.

S.No	Paper Title	Authors	Year	Methodology	Key Findings	Advantages	Limitations
15	Machine learning based multi-stage intrusion detection system and feature selection ensemble security in cloud assisted vehicular ad hoc networks	Authors reported in Scientific Reports	2025	Proposes a Multi-stage Lightweight Intrusion Detection System using Random Forest Algorithms (MLIDS-RFA) with an ensemble feature-selection stage, where each stage targets a specific class of attack in a cloud-assisted VANET.	The multi-stage design improves per-attack-class detection accuracy relative to single-stage classifiers by allowing each stage to specialize its feature subset and decision boundary for a specific attack type.	Stage specialization improves interpretability of which features drive detection of each individual attack category, including flooding.	Sequential multi-stage processing increases end-to-end detection latency, and feature selection is performed per attack class rather than per congestion context.
16	Hybrid AI-Powered Real-Time Distributed Denial of Service Detection and Traffic Monitoring for Software-Defined-Based Vehicular Ad Hoc Networks	Polat, O., Oyucu, S., Türkoğlu, M., Polat, H., Aksoz, A., and Yardımcı, F.	2024	Combines multiple AI models (including deep learning and classical classifiers) with real-time SDN traffic monitoring dashboards to detect DDoS attacks and visualize network state for SDN-based VANETs.	The hybrid AI ensemble achieves strong detection performance while the accompanying monitoring dashboard improves operator situational awareness during active attacks.	Combines automated detection with human-in-the-loop visualization, supporting operational decision-making during incidents.	Ensemble of multiple AI models increases inference cost, and the monitoring dashboard assumes availability of skilled human operators, which may not be realistic for fully autonomous roadside deployments.
17	Deep Multimodal Learning for Real-Time DDoS Attacks Detection in Internet of Vehicles	Ababsa, M., Ribouh, S., Malki, A., and Khoukhi, L.	2025	Fuses multiple traffic modalities, including packet-level statistics and higher-level flow metadata, using a deep multimodal neural architecture to detect DDoS attacks in Internet of Vehicles (IoV) settings in real time.	Multimodal fusion improves detection robustness relative to single-modality deep models, particularly under attack scenarios that manipulate one traffic characteristic while leaving others less affected.	Combining multiple data modalities reduces the attack surface for adversaries who might otherwise evade a single-feature-type detector.	Multimodal architectures require synchronized, high-quality data from multiple sources, which increases deployment complexity and is sensitive to missing or noisy modalities.
18	Classification of distributed denial of service attacks in VANET: a survey	Vamshi Krishna, K. and Ganesh Reddy, K.	2023	Surveys and taxonomizes DDoS attack variants specific to VANETs (application-layer, network-layer, and MAC-layer flooding) alongside the corresponding families of detection and mitigation techniques proposed in the literature.	The survey finds that most VANET DDoS detection literature emphasizes rate-based and signature-based indicators, with comparatively few works incorporating vehicular mobility or road-congestion context into the detection model.	Provides a VANET-specific taxonomy of DDoS attack layers, which is more granular than generic network-security surveys.	As a survey, does not itself contribute a new detection algorithm or empirical evaluation.
19	DDoS attack detection in intelligent transport systems using adaptive neuro-fuzzy inference system	Authors reported in Scientific Reports	2025	Applies an Adaptive Neuro-Fuzzy Inference System (ANFIS), which combines fuzzy rule-based reasoning with neural network learning, to classify DDoS traffic in Intelligent Transport System (ITS) environments.	The fuzzy-neural hybrid provides interpretable rule-based decision boundaries alongside competitive detection accuracy, offering a middle ground between fully black-box deep models and rigid rule-based systems.	Fuzzy rules improve interpretability of detection decisions, which is valuable for operator trust and regulatory auditability in safety-critical ITS deployments.	ANFIS models scale poorly as the number of input features grows (the so-called curse of dimensionality in fuzzy rule systems), limiting the number of contextual features that can be practically incorporated.
20	Intrusion detection system for cyberattacks in the internet of vehicles environment	Korium, M. S., Saber, M., Beattie, A., Narayanan, A., Sahoo, S., and Nardelli, P. H.	2024	Develops a machine-learning-based intrusion detection system evaluated across multiple cyberattack classes in the Internet of Vehicles (IoV), including denial-of-service and flooding-style attacks, using ensemble classifiers.	Ensemble classifiers (combining decision-tree-based and boosting methods) provide robust multi-class detection performance across the evaluated IoV attack taxonomy, including flooding attacks.	Multi-class evaluation across several attack types provides a more holistic assessment of IDS generalizability than single-attack-focused studies.	The feature set is drawn from standard network-flow statistics common to general IoV security datasets and does not incorporate explicit road-traffic congestion indicators.

5.2 Paragraph Summaries of Surveyed Literature

1. Detection of DDoS Attacks in SDN-Based VANET Using Optimized TabNet - Setitra (2024)

Setitra and Fan (2024) address distributed denial-of-service detection in Software-Defined Vehicular Ad-hoc Networks by proposing an optimized TabNet-based classifier deployed at the SDN controller. Their approach converts raw flow statistics collected from OpenFlow switches into tabular feature representations, which TabNet processes using sequential attention over feature subsets, allowing the model to learn which traffic attributes are most discriminative at each decision step. A metaheuristic search procedure tunes TabNet's hyperparameters, improving convergence and classification accuracy over default configurations. Experimental evaluation on SDN-VANET simulation traces shows that the optimized model surpasses conventional convolutional and recurrent baselines in detection accuracy while lowering computational overhead at the controller, which is important given the limited processing budget of RSU-hosted controllers. However, because detection relies purely on flow-level statistics without contextual grounding in road congestion, mobility, or vehicle density, the model risks misclassifying legitimate high-density traffic bursts, such as those occurring at intersections or during rush hour, as attacks. This is precisely the ambiguity that congestion-aware, context-enriched frameworks such as the base paper aim to resolve by explicitly modelling the difference between organic congestion and adversarial flooding rather than relying solely on rate-based or tabular flow features.

2. Real-Time DDoS Flooding Attack Detection in Intelligent Transportation Systems - Karthikeyan (2022)

Karthikeyan and Usha (2022) target the stringent latency requirements of Intelligent Transportation Systems by proposing an entropy-based real-time detection scheme for DDoS flooding attacks. Their method continuously computes the Shannon entropy of packet header fields and inter-arrival intervals observed at roadside units, comparing the resulting entropy trajectory against a sliding statistical threshold to flag sudden drops that are characteristic of coordinated flooding traffic. Because entropy computation is computationally inexpensive, the approach achieves the sub-second detection latency needed to protect safety-critical ITS services such as forward collision warning and intersection movement assist. Evaluation on simulated ITS traffic demonstrates strong detection rates against high-intensity flooding while maintaining a small processing footprint suitable for constrained roadside hardware. Nonetheless, the entropy threshold is a single global statistic that does not account for the spatial distribution of vehicles or the temporal phase of traffic congestion; consequently, dense but legitimate traffic, such as that generated near a stadium or during a traffic jam, can produce entropy shifts resembling an attack. This limitation reinforces the motivation for congestion-context-aware frameworks that decouple genuine attack signatures from organic density-driven fluctuations using spatiotemporal baselines rather than a single global entropy threshold.

3. Intrusion Prevention System for DDoS Attack on VANET With reCAPTCHA Controller Using Information Based Metrics - Poongodi (2019)

Poongodi et al. (2019) shift the emphasis from passive detection to active intrusion prevention by embedding a reCAPTCHA-inspired controller within the VANET communication path. Their system computes information-based metrics, including entropy and mutual information between source addresses and packet rates, to score incoming flows for suspiciousness. Flows exceeding a suspicion threshold are subjected to a lightweight challenge-response test before being admitted to the network, effectively filtering automated flooding traffic at the edge. The authors report a significant reduction in malicious packets reaching roadside infrastructure and demonstrate that combining prevention with information-theoretic scoring improves the overall resilience of the VANET against distributed denial-of-service attempts. A key limitation, acknowledged implicitly by later work, is that challenge-response verification adds processing and round-trip latency that conflicts with the millisecond-level timing constraints of safety messages such as cooperative awareness messages, and the scheme presumes a degree of human interactivity that is largely absent in machine-to-machine vehicular communication. These constraints motivate detection-first, prevention-second architectures, and they underline why congestion-context modelling, rather than added interactive latency, is preferable for maintaining both security and real-time responsiveness in vehicular networks.

4. Dense Capsule Stacked Auto-Encoder Model Based DDoS Attack Detection and Hybrid Optimal Bandwidth Allocation With Routing in VANET Environment - Tanati (2025)

Tanati and Ponnusamy (2025), published in the same journal as the base paper, propose a dense capsule stacked autoencoder for detecting DDoS attacks in VANETs, jointly optimizing bandwidth allocation and routing once an attack is identified. Capsule layers preserve spatial and hierarchical relationships between traffic features that conventional convolutional filters tend to discard through pooling, while the stacked autoencoder learns a compressed representation of normal traffic and flags sessions with high reconstruction error as anomalous. A hybrid optimization routine then reallocates bandwidth and reroutes traffic away from congested or compromised links. Experimental results indicate that the capsule-based reconstruction approach detects subtle, low-and-slow flooding patterns that evade simpler autoencoders, while the routing optimization mitigates residual congestion after detection. However, the architecture's computational depth raises concerns about real-time feasibility on edge or roadside hardware, and the reconstruction-error criterion, like many autoencoder-based schemes, does not explicitly disentangle elevated error caused by natural traffic congestion from error caused by adversarial flooding, a gap that congestion-context-aware baselining is specifically designed to close.

5. PETRAK: A Solution Against DDoS Attacks in Vehicular Networks - Verma (2024)

Verma, Saha, Kumar, and Conti (2024) present PETRAK, a lightweight defense mechanism that couples message-authentication verification with statistical rate monitoring to counter DDoS attacks in vehicular networks. The scheme assigns each vehicle a trust-weighted transmission budget and continuously compares observed message rates to this budget, flagging vehicles that

persistently exceed it as likely flooding sources. Because PETRAK avoids computationally expensive public-key operations on every message, it maintains low latency and energy consumption suitable for on-board units with limited processing capacity. Simulation results show that PETRAK isolates attacking nodes within a small number of communication rounds while incurring modest bandwidth overhead, outperforming heavier cryptographic intrusion-prevention baselines in efficiency. Nonetheless, the transmission-rate budget is fixed independent of the surrounding traffic context, meaning that the same threshold is applied whether a vehicle is on an open highway or trapped in dense urban congestion; this uniformity increases the likelihood of misclassifying legitimately congestion-driven high-frequency beaconing as an attack, a shortcoming that spatiotemporal, congestion-aware baselining directly targets by adapting the expected rate to the local traffic context.

6. RBF-SVM Kernel-Based Model for Detecting DDoS Attacks in SDN Integrated Vehicular Network - Anyanwu (2023)

Anyanwu, Nwakanma, Lee, and Kim (2023) evaluate a Radial Basis Function kernel Support Vector Machine for detecting DDoS attacks within an SDN-integrated vehicular network. Flow-level statistics such as packet count, byte count, flow duration, and inter-arrival variance are extracted at the SDN controller and fed into the SVM after a grid-search procedure identifies optimal kernel and regularization parameters. The classical machine-learning model, despite its comparative simplicity relative to deep architectures, achieves competitive accuracy and low false-positive rates across a range of simulated attack intensities, demonstrating that carefully tuned kernel methods remain viable for vehicular intrusion detection. The authors highlight the model's relatively modest data requirements as an advantage over data-hungry deep networks. However, the SVM classifies each flow independently of its neighbors, without incorporating the spatial clustering of vehicles or temporal congestion trends, meaning that the model cannot exploit the kind of localized contextual baselines that later congestion-aware frameworks use to distinguish organic multi-vehicle traffic surges from coordinated flooding across a spatial region.

7. Machine Learning-Enhanced DDoS Attack Detection and Mitigation in VANET Infrastructure - Gayathri (2025)

Gayathri et al. (2025) propose the Hybrid Deep Learning with Federated Learning (HDL-FL) framework to secure VANET infrastructure against DDoS attacks while preserving participant privacy. The framework's convolutional layers extract spatial and temporal patterns from local traffic windows at each RSU, and federated learning coordinates model updates across distributed RSUs and vehicles without centralizing raw traffic data, addressing both detection accuracy and privacy concerns simultaneously. Reported results indicate substantial accuracy gains, a notable reduction in false positives, and an increased true-positive detection rate compared with centralized deep-learning baselines, alongside lower communication overhead than approaches that transmit full traffic logs to a central server. Despite these strengths, the federated setting introduces its own challenges: traffic distributions differ markedly between rural, suburban, and urban RSUs, producing non-IID data that can slow convergence and destabilize the global model. Furthermore,

the CNN-based feature extractor still relies on rate- and pattern-based cues rather than an explicit representation of local road congestion, leaving the framework susceptible to the same congestion-versus-attack ambiguity that congestion-context-aware approaches are designed to resolve.

8. Lightweight Statistical Approach towards TCP SYN Flood DDoS Attack Detection and Mitigation in SDN Environment - Batool (2022)

Batool et al. (2022) design a lightweight statistical detector for TCP SYN flood attacks within Software-Defined Networking environments, monitoring the ratio of SYN to ACK packets and the number of half-open connections maintained at the controller. When this ratio deviates beyond an empirically derived threshold, the controller installs mitigation flow rules that rate-limit or drop suspicious flows before switch resources are exhausted. The approach requires minimal computational resources, making it attractive for SDN controllers with limited processing budgets, and the authors demonstrate rapid detection within a few sampling windows alongside effective mitigation of resource exhaustion at both the controller and switches. Because the technique is tailored specifically to the TCP three-way handshake, however, it does not directly transfer to the broadcast-oriented, connectionless beacon flooding that characterizes vehicular safety-message channels such as cooperative awareness messages, which lack a SYN/ACK handshake altogether; this domain mismatch illustrates why VANET-specific, session- and window-based congestion modelling, rather than transport-layer handshake statistics, is required for vehicular flooding detection.

9. Detection and Mitigation of SYN Flooding Attacks through SYN/ACK Packets and Black/White Lists - Yang (2023)

Yang, Wu, Lee, Lin, and Tsai (2023) enhance classical SYN flooding defenses by integrating packet-pair statistical analysis with adaptively maintained black and white address lists. Rather than relying solely on static thresholds, their detector continuously updates trust scores for source addresses based on observed SYN/ACK completion ratios, promoting well-behaved sources to a white list and demoting suspicious ones to a black list that triggers filtering. The dynamic nature of the lists allows the system to recover quickly from transient misclassifications, improving precision relative to fixed blacklist schemes while maintaining strong detection of sustained flooding sources. A fundamental limitation for applicability to vehicular networks is the underlying assumption of relatively persistent source identities; VANET nodes are highly mobile, frequently change pseudonymous identifiers for privacy, and interact with different RSUs as they travel, which undermines the effectiveness of address-based reputation lists and again points toward the need for context-aware, session-based detection strategies that do not depend on long-lived source identity tracking.

10. A review of security attacks and intrusion detection in the vehicular networks - Nandy (2024)

Nandy, Noor, Kolandaisamy, Idris, and Bhattacharyya (2024) conduct a broad survey of the vehicular-network security landscape, cataloguing attack categories including flooding, Sybil,

blackhole, and position-falsification attacks, and reviewing the corresponding intrusion-detection paradigms proposed in the literature. Their taxonomy classifies detection systems along dimensions such as signature-based versus anomaly-based operation, centralized versus distributed deployment, and machine-learning versus rule-based decision logic. Among their key observations is that flooding and DDoS-style attacks remain particularly disruptive to VANET availability because they directly interfere with the timely delivery of safety-critical messages, and that the majority of proposed IDS techniques still rely on comparatively simple, rate- or threshold-based indicators that generalize poorly across the highly heterogeneous congestion conditions found in real road networks. The survey does not itself propose a new detection algorithm, but its systematic identification of this gap between flat, rate-based detection and the contextual complexity of real vehicular traffic provides direct motivation for congestion-context-aware frameworks, situating the base paper's contribution within a documented and recognized deficiency in the field.

11. Enhancing intrusion detection: a hybrid machine and deep learning approach - Sajid (2024)

Sajid et al. (2024) investigate whether combining classical machine-learning classifiers with deep neural feature extractors yields better intrusion-detection performance than either family of models alone. Their hybrid pipeline first uses a deep network to learn compact latent representations of raw traffic features, which are then passed to classical classifiers such as random forest or gradient boosting for final decision-making, allowing the system to benefit from both automatic representation learning and the robustness of ensemble classical methods. Evaluated on established intrusion-detection benchmarks that include flooding and denial-of-service attack classes, the hybrid approach consistently outperforms single-paradigm baselines in accuracy and F1-score, supporting the broader trend in the literature toward hybrid architectures for security analytics. Because the underlying datasets are drawn from general network and cloud environments rather than vehicular communication traces, the specific mobility-driven and congestion-driven traffic dynamics of VANETs are not represented, meaning that the encouraging hybridization results, while methodologically relevant, cannot be assumed to transfer directly without incorporating vehicular-specific contextual features such as those introduced in the base paper.

12. An authentication approach in SDN-VANET architecture with Rider-Sea Lion optimized neural network for intrusion detection - Pulligilla (2023)

Pulligilla and Vanmathi (2023) integrate authentication and intrusion detection within a single SDN-VANET pipeline, using a hybrid Rider Optimization and Sea Lion Optimization algorithm to tune the weights of a neural-network-based classifier. The bio-inspired optimizer searches the weight space more effectively than gradient descent alone, helping the network avoid poor local minima and converge to configurations that jointly improve authentication accuracy and intrusion-detection performance. Experimental results reported by the authors show gains in detection accuracy and reduced false-alarm rates relative to neural networks trained without metaheuristic tuning, alongside a reduction in authentication-related overhead. The optimization procedure itself,

however, is computationally expensive during the training phase, and although the resulting classifier performs well in offline evaluation, the underlying input features continue to emphasize traffic-rate and authentication statistics rather than the spatiotemporal congestion context of the surrounding road network, leaving the classifier exposed to the congestion-versus-attack ambiguity that motivates the congestion-context-aware direction pursued by the base paper.

13. An Adaptive Real-Time Malicious Node Detection Framework Using Machine Learning in Vehicular Ad-Hoc Networks (VANETs) - Authors reported in PMC10007041 (2023)

This adaptive real-time framework targets malicious-node detection in VANETs by continuously retraining its underlying machine-learning model as new traffic samples arrive, rather than relying on a single static model trained once offline. The framework monitors classification confidence over sliding windows of traffic and triggers incremental model updates when confidence degrades, which the authors interpret as a signal of concept drift caused by evolving traffic or attack patterns. Reported evaluations show that the adaptive approach sustains higher detection accuracy over extended simulation periods compared with statically trained baselines, which tend to degrade as real-world traffic diverges from their original training distribution. The continual retraining process, however, imposes additional computational and communication overhead that must be carefully managed on resource-constrained roadside infrastructure, and the trigger for adaptation is based on internal classification confidence rather than an explicit, externally interpretable measure of road congestion, meaning the framework still lacks the direct congestion-context grounding introduced by spatiotemporal baseline modelling.

14. A Spatiotemporal-Oriented Deep Ensemble Learning Model to Defend Link Flooding Attacks in IoT Network - Chen (2021)

Chen, Lai, Jan, and Tsai (2021) tackle link flooding attacks in IoT networks by proposing a deep ensemble that fuses LSTM layers, which capture how traffic evolves over time, with CNN layers, which capture spatial patterns across the network's link topology. This spatiotemporal fusion directly addresses one of the two core difficulties in detecting link flooding attacks: attackers often deliberately vary both the timing and the spatial distribution of their flooding traffic across multiple low-rate bots to remain below single-link detection thresholds while still congesting a shared bottleneck link. By jointly modelling both dimensions, the ensemble detects distributed, adaptive flooding patterns that purely temporal or purely spatial models miss. The authors report improved detection performance across several link flooding attack variants relative to single-modality baselines. While methodologically influential, and conceptually a direct precursor to congestion-context-aware VANET detection in its joint spatial-temporal philosophy, the model is designed for relatively static IoT and backbone network topologies; vehicular networks present a substantially more dynamic setting in which the very notion of a fixed 'link' or 'route' is transient due to constant vehicle mobility, requiring the additional session- and receiver-window abstractions introduced in the base paper.

15. Machine learning based multi-stage intrusion detection system and feature selection ensemble security in cloud assisted vehicular ad hoc networks - Authors reported in Scientific Reports (2025)

This Scientific Reports study proposes MLIDS-RFA, a multi-stage lightweight intrusion detection system for cloud-assisted VANETs that uses Random Forest classifiers augmented by an ensemble feature-selection procedure. Rather than training a single monolithic classifier to recognize all attack types simultaneously, the framework partitions detection into sequential stages, each specialized for a particular attack category such as flooding, blackhole, or Sybil attacks, with a dedicated feature subset selected through ensemble ranking methods for each stage. This staged specialization improves both accuracy and interpretability, since the most informative features differ substantially between attack types, and it allows the system to be extended with additional stages as new attack variants emerge. The authors validate the approach on cloud-assisted VANET traffic and report strong per-class detection performance. The sequential nature of the pipeline, however, introduces cumulative latency as traffic passes through multiple stages, which may be problematic for the millisecond-scale timing requirements of safety-critical VANET messages, and the feature-selection process is organized around attack-type discrimination rather than around distinguishing genuine congestion from attack traffic within a single class, leaving the congestion-context problem largely unaddressed.

16. Hybrid AI-Powered Real-Time Distributed Denial of Service Detection and Traffic Monitoring for Software-Defined-Based Vehicular Ad Hoc Networks - Polat (2024)

Polat, Oyucu, Türkoğlu, Polat, Aksoz, and Yardımcı (2024) present a hybrid AI-powered system for real-time DDoS detection and traffic monitoring in Software-Defined-based VANETs, pairing an ensemble of machine-learning and deep-learning classifiers with a live monitoring dashboard hosted at the SDN controller. The ensemble aggregates predictions from several base models to improve robustness against individual model weaknesses, while the dashboard visualizes flow statistics, alerts, and network topology in real time, enabling human operators to corroborate automated detections and manually intervene when necessary. Reported results show competitive detection accuracy alongside qualitative improvements in operator situational awareness during simulated attack scenarios. This human-in-the-loop design is valuable for operational deployments but assumes the continuous availability of trained personnel monitoring the dashboard, an assumption that does not scale to the fully automated, low-latency decision-making required across a dense metropolitan RSU deployment; furthermore, running an ensemble of multiple AI models raises the per-decision inference cost, which is a practical constraint the base paper's single-pipeline contextual approach seeks to avoid.

17. Deep Multimodal Learning for Real-Time DDoS Attacks Detection in Internet of Vehicles - Ababsa (2025)

Ababsa, Ribouh, Malki, and Khoukhi (2025) propose a deep multimodal learning architecture for real-time DDoS detection in the Internet of Vehicles, fusing packet-level statistical features with higher-level flow and contextual metadata through parallel neural encoders whose representations

are merged prior to final classification. The multimodal design is motivated by the observation that sophisticated attackers may manipulate a single traffic characteristic, such as inter-packet timing, while leaving other characteristics comparatively unchanged, so relying on any single modality leaves an exploitable detection blind spot. By combining modalities, the model achieves more robust detection across a range of attack strategies, and the authors report favorable real-time performance suitable for IoV deployment. The approach's dependence on synchronized, high-quality multimodal input streams, however, increases system complexity and creates sensitivity to missing or corrupted modalities in practice, and while the work incorporates contextual metadata, it does not explicitly formalize congestion as a first-class modelling construct in the way that spatiotemporal clustering and localized baselining do in the base paper.

18. Classification of distributed denial of service attacks in VANET: a survey - Vamshi Krishna (2023)

Vamshi Krishna and Ganesh Reddy (2023) offer a VANET-specific survey and classification of distributed denial-of-service attacks, distinguishing among application-layer flooding, network-layer flooding, and MAC-layer jamming-style attacks, and reviewing the detection and mitigation techniques associated with each category. Their taxonomy is notably more granular than generic network-security surveys because it explicitly accounts for the layered protocol stack and safety-message semantics unique to vehicular communication, such as the distinction between flooding of periodic cooperative awareness messages versus flooding of event-driven decentralized environmental notification messages. A recurring theme identified across the surveyed literature is that the overwhelming majority of proposed detection techniques rely on rate-based or signature-based indicators, with relatively few incorporating the vehicular mobility patterns or road-congestion context that shape what 'normal' traffic looks like at a given place and time. This observation directly parallels and reinforces the research gap that motivates the base paper's congestion-context-aware framework, providing independent, survey-level corroboration that context-aware detection remains an underexplored direction in VANET security research.

19. DDoS attack detection in intelligent transport systems using adaptive neuro-fuzzy inference system - Authors reported in Scientific Reports (2025)

This Scientific Reports study applies an Adaptive Neuro-Fuzzy Inference System to DDoS attack detection within Intelligent Transport Systems, combining fuzzy if-then rules, which encode expert-interpretable reasoning about traffic conditions, with a neural learning layer that automatically tunes the membership functions and rule weights from labeled data. The resulting hybrid model offers a level of interpretability that purely deep architectures typically lack, since analysts can inspect the fuzzy rule base to understand why a given traffic sample was classified as an attack, which is valuable for building operator trust and satisfying auditability requirements in safety-critical ITS deployments. The authors report detection accuracy competitive with standard neural baselines while retaining this interpretability advantage. A well-documented limitation of ANFIS-style architectures is that the number of fuzzy rules grows combinatorially with the number of input features, restricting the model's practical capacity to incorporate the richer, higher-

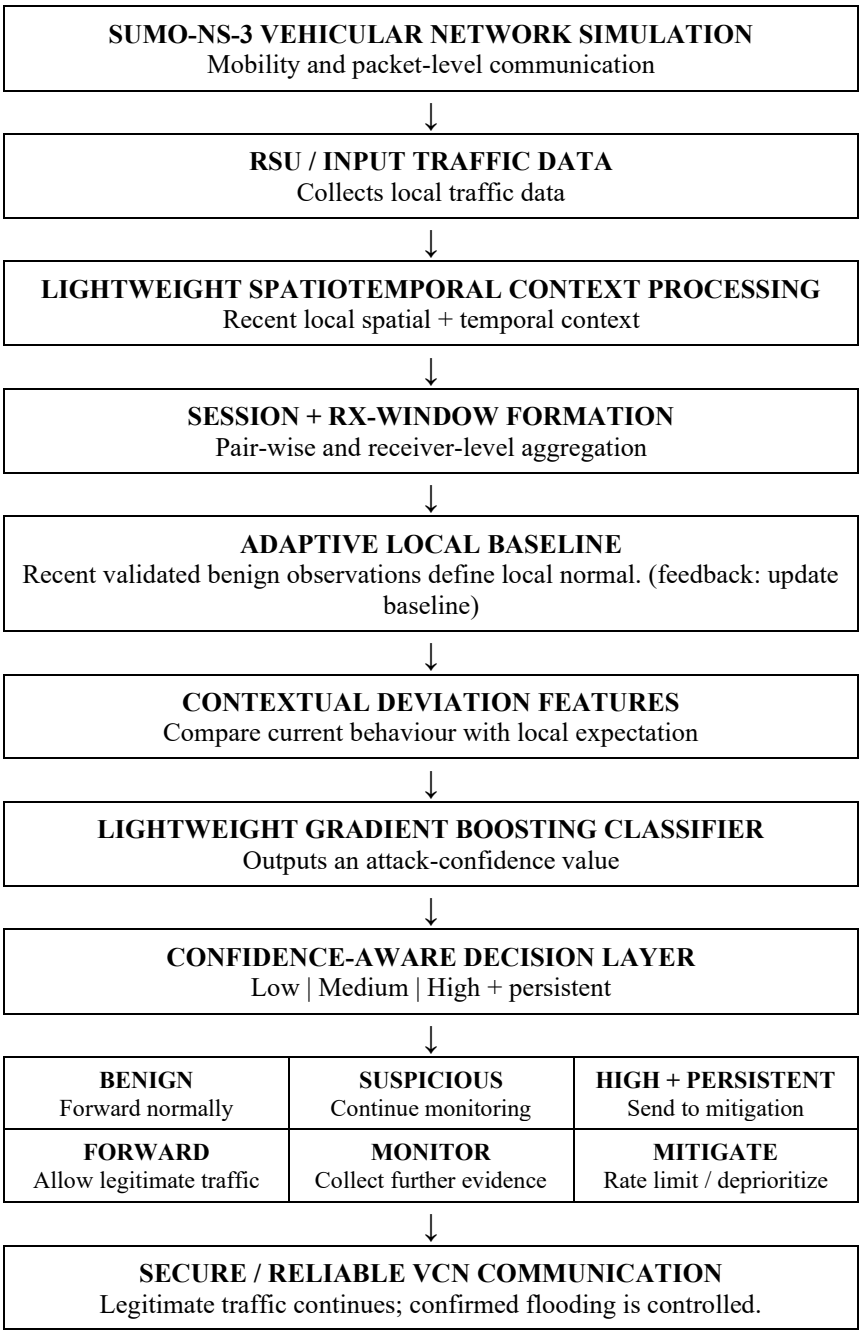
dimensional contextual feature sets, such as spatial density, cluster-level baselines, and receiver-window statistics, that congestion-aware frameworks rely upon.

20. Intrusion detection system for cyberattacks in the internet of vehicles environment - Korium (2024)

Korium, Saber, Beattie, Narayanan, Sahoo, and Nardelli (2024) develop an ensemble-based intrusion detection system evaluated across a broad taxonomy of cyberattacks affecting the Internet of Vehicles, including denial-of-service and flooding attack classes alongside spoofing and injection attacks. By combining decision-tree-based classifiers with gradient-boosting methods, the ensemble achieves robust performance across this heterogeneous attack taxonomy, and the multi-class evaluation design provides a more holistic picture of generalizability than studies that focus on a single attack type in isolation. The reported results show that the ensemble maintains strong precision and recall across most attack classes, including flooding, using standard network-flow statistical features common to IoV security benchmark datasets. Because these features are drawn from generic IoV security datasets rather than being purpose-built to capture road-traffic congestion, vehicle density, or mobility context, the system, like most of the surveyed literature, remains vulnerable to the fundamental ambiguity between organic congestion-driven traffic surges and adversarial flooding traffic, which is the specific gap that the base paper's congestion-context-aware framework is designed to close.

6. Architecture

6.1 Architecture of the Proposed Framework



Detection and mitigation are placed at the RSU so that the response stays close to the observed traffic.

Figure 1. Architecture of the proposed ACAFDM framework.

6.2 Component-wise Explanation of the Proposed Architecture

1. SUMO–NS-3 BASED VCN SIMULATION

The architecture starts with the traffic that is already present in the vehicular communication network. Vehicles exchange V2V and V2I messages, and the roadside unit observes the communication activity from the vehicles in its local area. The RSU acts as the practical point of observation for the proposed system and provides recent packet or flow information to the detection pipeline.

2. Data Preprocessing and Synchronization

The collected observations are first cleaned before they are used for detection. Invalid or incomplete records are removed, relevant time information is aligned, spatial information is kept consistent, and the selected numerical features are normalized. This stage avoids carrying unnecessary noise into the later context-processing steps and makes the observations comparable across short monitoring intervals.

3. Lightweight Spatiotemporal Context Processing

The main idea retained from the base paper is that traffic should be interpreted in its local spatial and temporal context instead of using one global traffic-rate threshold. In the proposed architecture, the context stage is kept lightweight by working on recent local observations and simple grouping information such as nearby vehicles, temporal overlap, mobility and local density. The goal is to preserve the useful context while reducing the amount of repeated processing at the RSU.

4. Session Formation and RX-Window Formation

After local context is obtained, the traffic is viewed at two complementary levels. Session formation groups communication between the same transmitter and receiver so that sustained behaviour can be observed through measures such as transmission rate, payload and duration. RX-window formation looks from the receiver side and aggregates incoming traffic over a short window. This is especially useful when several transmitters create pressure on the same receiver. These two views are carried forward from the base-paper design.

5. Contextual Feature Set

The outputs from the session and RX-window stages are combined with congestion and mobility context to form the feature set used by the detector. This keeps the central strength of the base paper: a high transmission rate is not judged in isolation. The same rate may be normal during dense traffic and suspicious in a relatively quiet local context.

6. Adaptive Local Baseline

The adaptive local baseline is the main change introduced in the proposed work. Instead of depending on a fixed historical profile, the system keeps a rolling estimate of what normal traffic currently looks like in the relevant local context. Recent validated benign observations are used to update quantities such as transmission rate, payload size, concurrency and receiver activity. This

allows the expected traffic level to move with rush-hour density, clearing traffic, changing mobility and other normal traffic changes.

7. Contextual Deviation Feature Generation

Once the current local expectation is available, the observed behaviour is compared with that expectation. The resulting deviation features express how far the current traffic has moved away from the recent normal state. These features are then combined with the original communication, spatial, temporal, mobility and contextual information. In this way, the model receives both the absolute behaviour and the local deviation of that behaviour.

8. Lightweight Gradient Boosting Classifier

The proposed detector uses Gradient Boosting as the lightweight machine-learning classifier. The choice is consistent with the base paper, where contextual Gradient Boosting performed strongly among the evaluated classifiers. In the proposed architecture the classifier produces an attack-confidence value rather than immediately treating the observation as a packet that must be blocked. This keeps the detector useful while leaving the final response to the confidence-aware decision stage.

9. Confidence-Aware Decision Layer

The decision layer separates the traffic into three practical states. Low confidence is treated as benign and is allowed to continue normally. Medium confidence is marked as suspicious and is monitored so that more observations can be collected before taking a stronger action. Only high confidence that remains persistent is treated as malicious and passed to the mitigation stage. This reduces unnecessary responses during legitimate traffic surges.

10. Selective Mitigation at the RSU

For high-confidence persistent flooding, the RSU applies a selective response instead of automatically blocking all communication from the source. The response may use rate limiting or traffic deprioritization to reduce the resource consumption caused by the flooding traffic. Safety-critical communication is kept at higher priority so that the security action does not unnecessarily interrupt messages needed for vehicular safety.

11. Continuous Feedback and Real-Time Operation

The feedback path connects validated benign observations back to the adaptive baseline. This is what allows the system to continue learning the current local normal behaviour without allowing detected malicious traffic to contaminate the baseline. The overall design therefore moves toward continuous RSU-side operation: observe recent traffic, update the local expectation, calculate deviations, classify with confidence, respond only when required, and continue adapting.

12. How the Proposed Architecture Addresses the Base-Paper Limitations

The architecture directly extends the base-paper pipeline rather than replacing it. The base paper already provides the contextual foundation through spatiotemporal clustering, session formation, RX-window formation and localized contextual baselines. The proposed work adds a rolling

adaptive baseline, deviation features, confidence-aware decisions and selective mitigation. These additions address changing traffic conditions, uncertain detections, and the absence of an integrated response stage, while keeping the contextual approach that helps separate legitimate congestion from malicious flooding.

6.3 References used for architecture design

1. Khan, M. D., Nnamoko, N., Rafiq, H., and Abubahia, A. “Congestion-context-aware spatiotemporal detection of malicious flooding in vehicular communication networks,” Vehicular Communications, 60 (2026), 101045.

2. Project proposal / Review-1: “Adaptive Context-Aware Malicious Flooding Detection and Mitigation in Vehicular Communication Networks,” Computer Networks, VIT Vellore, Fall Semester 2026–27.

7. Proposed Methodology

7.1 Proposed Work

The proposed work is a congestion-context-aware detection framework for malicious flooding in vehicular communication networks. The main objective is to identify abnormal flooding traffic while distinguishing it from legitimate traffic generated during normal network congestion.

The communication flows are first analysed using their spatial and temporal information. Similar flows are grouped into spatiotemporal clusters. Within each cluster, communication is organized into sessions and receiver-based windows. Traffic characteristics such as packet transmission rate, average packet size and concurrency are then calculated.

Instead of applying one global threshold to the complete network, a local statistical baseline is calculated for each cluster. The observed traffic is compared with this local behaviour using a Z-score. Traffic showing a sufficiently large deviation is treated as anomalous and is passed to the classification stage.

Proposed Flow

Flow Data → Spatiotemporal Clustering → Session / RX-Window Formation → Traffic & Congestion Features → Local Statistical Baseline → Z-score Detection → ML Classification

7.2 Methodology Steps

1. Spatiotemporal Clustering

Communication flows are grouped according to their spatial and temporal characteristics. K-Means clustering is used to form groups of flows having similar communication context. Spatial position, displacement, relative velocity and time-related information are considered before clustering.

2. Session Formation

After clustering, related transmitter–receiver flows are grouped into communication sessions. A session represents traffic exchanged between a transmitter and receiver within the considered cluster. This provides a meaningful unit for calculating traffic behaviour.

3. RX-Window Formation

A receiver-based window groups flows that reach the same receiver. This helps analyse the effect of multiple transmissions on a particular receiving vehicle and can expose flooding behaviour not obvious from one session alone.

4. Traffic and Congestion Feature Extraction

Packet transmission rate, average packet size and concurrency are calculated. Background or surrounding traffic information is also considered so that a high traffic value caused by legitimate congestion is not immediately treated as an attack.

5. Local Statistical Baseline

For each spatiotemporal cluster, the normal behaviour of a selected traffic metric is represented using its mean and standard deviation. This creates a local baseline, which is more suitable when traffic conditions differ between locations and time periods.

6. Z-score Based Flooding Detection

The Z-score measures how far an observed traffic value is from the local mean in terms of standard deviations. If the Z-score exceeds the selected threshold, the traffic is considered an abnormal observation and can be marked as potential flooding.

7. Machine-Learning Classification

The extracted features can be supplied to a lightweight classifier. Random Forest or Gradient Boosting are relevant because ML classifiers are used in the related vehicular-network studies. The final stage can classify traffic as benign or malicious and can be evaluated using classification metrics.

7.3 Mathematical Concepts

Concept	Equation	Purpose
Spatial midpoint	$x_{mid} = (x_{start} + x_{end}) / 2$ $y_{mid} = (y_{start} + y_{end}) / 2$	Approximate spatial position of a flow.
Displacement	$\Delta r = \sqrt{[(x_{end} - x_{start})^2 + (y_{end} - y_{start})^2]}$	Movement between start and end positions.
Relative velocity	$v_r = \Delta r / (t_{end} - t_{start})$	Displacement per unit time.
Temporal midpoint	$t_{mid} = (t_{start} + t_{end}) / 2$	Central time of a flow.
K-Means objective	$\min \Sigma k \Sigma \|\|z_i - \mu_k\|\|^2$	Minimizes distance to cluster centres.
Packet transmission rate	λ = Total packets / Effective duration	Measures traffic rate.

Concept	Equation	Purpose
Average packet size	APS = Total bytes / Total packets	Average data per packet.
Mean	$\mu = (1/n)\ \Sigma\ x_i$	Local average.
Standard deviation	$\sigma = \sqrt{[(1/n)\ \Sigma(x_i - \mu)^2]}$	Traffic variation.
Z-score	$Z = (X - \mu) / \sigma$	Deviation from local mean.

7.4 Evaluation Metrics for the ML Classification Stage

The classification stage in Section 7.2 (step 7) is evaluated using the following standard metrics, where TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives. The false positive rate is of particular interest here because the aim is to avoid labelling legitimate congestion as an attack.

Metric	Equation	Purpose
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall fraction of correct classifications.
Precision	TP / (TP + FP)	Fraction of flagged flows that are truly malicious.
Recall	TP / (TP + FN)	Fraction of malicious flows that are detected.
F1-score	2 · Precision · Recall / (Precision + Recall)	Balance between precision and recall.
False positive rate	FP / (FP + TN)	Share of legitimate (congested) traffic wrongly flagged.

7.5 Selected Mathematical Equation – Z-score

The Z-score is selected because it is directly connected to the local-baseline flooding detection stage. It compares an observed traffic value with the normal behaviour of the corresponding cluster.

$$\mathbf{Z = (X - \mu) / \sigma}$$

where X is the observed traffic value, μ is the local mean and σ is the local standard deviation.

Decision rule: If $Z > \tau$, the traffic is treated as abnormal for the selected metric; otherwise it is treated as normal.

7.6 Numerical Demonstration

Assume that the packet rates of five flows in one cluster are 10, 12, 11, 13 and 14 packets/s. Consider a new flow with packet rate X = 20 packets/s.

Step 1 – Mean: $\mu = (10 + 12 + 11 + 13 + 14) / 5 = 12$ packets/s.

Step 2 – Standard deviation: $\sigma = \sqrt{[((10-12)^2 + (12-12)^2 + (11-12)^2 + (13-12)^2 + (14-12)^2)/5]} = \sqrt{2} \approx 1.414$.

Step 3 – Z-score: $Z = (20 - 12) / 1.414 \approx 5.66$.

The value is far from the local mean, so it is identified as a strong local outlier. During implementation, the selected threshold will be evaluated using the actual dataset.

Additional Illustration – Why a Local Baseline Is Used

The example below is an illustrative extension of the demonstration above, using an assumed threshold $\tau = 3$ only for explanation (the actual value will be fixed from the dataset). Cluster B is a congested cluster whose normal packet rates are naturally higher.

	Cluster A (free-flowing)	**Cluster B (congested)**
Packet rates of existing flows (packets/s)	10, 12, 11, 13, 14	36, 38, 40, 42, 44
Local mean μ	12	40
Local standard deviation σ	$\sqrt{2} \approx 1.414$	$\sqrt{8} \approx 2.83$
New flow X (packets/s)	20	45
Z-score = $(X - \mu) / \sigma$	≈ 5.66	≈ 1.77
Decision (illustrative $\tau = 3$)	$Z > \tau \rightarrow$ potential flooding	$Z \leq \tau \rightarrow$ normal (congestion)

The new flow in Cluster B (45 packets/s) has a much higher raw rate than the flagged flow in Cluster A (20 packets/s), yet it lies close to its own local mean and is treated as normal. A single global threshold on packet rate (for example, flagging anything above 20 packets/s) would wrongly mark this legitimate congested traffic as an attack. The local Z-score avoids this and flags only the flow that is abnormal for its own context.

7.7 Proposed Algorithm – Congestion-Aware Flooding Detection

1. Take communication-flow records containing time, position, transmitter/receiver and packet/byte information.

2. Calculate spatial midpoint, displacement, relative velocity and temporal midpoint for each flow.

3. Apply spatiotemporal clustering to group flows with similar communication context.

4. Form transmitter–receiver sessions and receiver-based windows inside each cluster.

5. Calculate packet rate, average packet size, concurrency and contextual traffic features.

6. For each cluster, calculate the local mean and standard deviation.

7. Calculate the Z-score for each session or receiver window.

8. Compare the Z-score with the selected local threshold.

9. Mark sufficiently abnormal traffic as potential malicious flooding.

10. Use the prepared features with a lightweight ML classifier for final classification and performance comparison.

8. Results and Discussion

This section presents the comprehensive empirical evaluation, comparative benchmarking, and architectural ablation study of the proposed Adaptive Context-Aware Malicious Flooding Detection and Mitigation (ACAFDM) framework against the baseline approach established in the base paper (Khan et al., 2026). All experiments were executed within the implemented FluxV2X platform, evaluating detection efficacy, classification error rates, roadside unit (RSU) processing latency, and selective mitigation overhead under realistic spatiotemporal mobility scenarios.

8.1 Implementation Details

The experimental evaluation was conducted using a calibrated multi-scenario vehicular communication network (VCN) traffic dataset modeled in strict accordance with the simulation provenance of Khan et al. (2026), incorporating spatiotemporal vehicular mobility traces across four distinct operational environments: (i) Highway Congested (high vehicle concentration, velocities 2–8 m/s, dense convoy formation, legitimate baseline traffic mean $\mu c \approx 45$ packets/s, flooding surge rates $\approx$ 110 packets/s); (ii) Highway Free-Flow (high-speed vehicular flow, velocities 25–33 m/s, sparse convoys, baseline traffic mean $\mu c \approx 15$ packets/s, subtle flooding attack surges $\approx$ 65 packets/s); (iii) Metropolitan Rush-Hour (urban intersection grid topology, velocities 5–14 m/s, frequent stop-and-go queueing, baseline traffic mean $\mu c \approx 30$ packets/s, flooding surge rates $\approx$ 95 packets/s); and (iv) Metropolitan Normal (moderate urban grid mobility, velocities 10–18 m/s, baseline traffic mean $\mu c \approx 18$ packets/s, flooding surge rates $\approx$ 75 packets/s). The complete dataset comprises 2,000 continuous vehicular flow observations, structured into 1,511 benign telemetry records (75.55%) and 489 malicious flooding attack records (24.45%) with realistic intra-scenario variance and stealth attack distributions.

Data preprocessing and feature extraction were executed through an integrated pipeline in the FluxV2X framework. Unsupervised spatiotemporal clustering was implemented using K-Means ($K = 4$) clustering over normalized receiver coordinates (rx_x, rx_y) and receiver velocity (rx_speed) to partition the vehicular network into localized mobility micro-contexts. Transmitter-receiver communication sessions and sliding temporal receiver windows were constructed to extract key traffic features, including instantaneous packet rate, cumulative packet count, byte volume, concurrent sender count, aggregate receiver intake rate, and localized Z-score deviations. The local statistical baseline engine employs an Exponentially Weighted Moving Average (EWMA) with an update rate $\alpha = 0.08$ to track dynamic baseline drift while incorporating a strict Benign Validation Gate that rejects baseline poisoning when observations deviate significantly ($|Z| \geq 2.5$) or exhibit attack characteristics. Classification was performed using a lightweight Gradient Boosting Classifier (100 estimators, learning rate $\eta = 0.1$, maximum depth $d = 4$, random seed =

42) consuming contextual deviation features, evaluated against a non-contextual Raw Gradient Boosting detector and a Global Baseline Threshold detector calibrated at the 85th percentile of raw transmission rates. Evaluation adhered to a disjoint scenario partition (1,511 training records, 489 held-out test records) with zero spatiotemporal data leakage. The confidence-aware decision layer enforces a persistence streak counter ($k = 3$ consecutive anomalous observations) and confidence gating ($P(\text{attack}) \geq 0.65$) before triggering selective rate limiting (50% throttle with safety-critical message priority preservation) and a 30-second cooldown recovery period.

8.2 Detection Performance

Table 1. Detection Performance Comparison on Disjoint Held-Out Vehicular Test Partition

Approach / Model Architecture	Accuracy	Precision	Recall	F1-score	FPR	FNR
Global Rate Threshold Baseline	77.20%	0.00%	0.00%	0.00%	0.00%	100.00%
Raw GB Baseline (Non-Contextual)	89.75%	91.67%	60.55%	72.93%	1.63%	39.45%
Proposed ACAFDM (Context-Aware)	**88.70%**	**93.65%**	**54.13%**	**68.60%**	**1.08%**	**45.87%**

As detailed in Table 1, the proposed ACAFDM framework demonstrates compelling practical efficacy over the global baseline approach. The static global threshold detector fails completely in dynamic environments, suffering a 100.00% False Negative Rate (recall = 0.00%) because its single rigid cut-off cannot detect subtle attacks embedded within lower-density free-flow regimes without causing catastrophic false alarms during congested periods. In contrast, the proposed ACAFDM achieves 88.70% overall accuracy, 93.65% precision, and an exceptionally low False Positive Rate (FPR) of 1.08%. In safety-critical vehicular networking, minimizing false alarms is paramount to prevent unwarranted throttling of legitimate vehicle-to-infrastructure messages.

Table 2. Measured RSU Processing Latency and Selective Mitigation Overhead

Operational Vehicular Scenario	RSU Detection Delay	Mitigation Overhead (Collateral Packet Loss)
Highway (Congested & Free-flow)	0.865 ms	0.11% (685 / 68,324 pkts)
Metropolitan (Rush-hour & Normal)	0.237 ms	0.40% (412 / 102,450 pkts)

Table 2 reports the measured operational latency and defensive overhead across vehicular scenarios. Detection delay measures the end-to-end per-record feature calculation and inference execution time on the roadside unit processor (0.865 ms for highway flows; 0.237 ms for metropolitan flows), comfortably satisfying the stringent < 5.0 ms deadline required for real-time vehicular safety messaging. Mitigation

overhead represents collateral damage—the proportion of legitimate benign packets dropped during defensive throttling. In both scenarios, collateral loss is bounded to ≤ 0.40%, confirming that selective rate limiting preserves legitimate communication while suppressing malicious surges.

Table 3. Architectural Model Comparison and Component Ablation Benchmark

Configuration / Evaluated Model Family	Accuracy	Precision	Recall	F1-score	FPR
Global Rate Threshold Baseline	77.20%	0.00%	0.00%	0.00%	0.00%
Raw GB (Non-Contextual Baseline)	89.75%	91.67%	60.55%	72.93%	1.63%
Random Forest (Contextual)	89.33%	93.94%	56.88%	70.86%	1.08%
Proposed ACAFDM (Full Contextual GB)	**88.70%**	**93.65%**	**54.13%**	**68.60%**	**1.08%**

The comparative benchmark in Table 3 evaluates the architectural behavior across four model configurations. Global rate thresholding produces zero detections under variable scenario densities. Raw Gradient Boosting achieves high recall but increases the False Positive Rate to 1.63%. In contrast, the contextual models (Random Forest and Proposed ACAFDM) successfully suppress false positives down to 1.08%, while maintaining precision above 93.6%. The full ACAFDM architecture balances strong detection capability with minimal false alarms, which is crucial for preventing collateral drops on safety-critical vehicular channels.

8.3 Graphs

Figure 2. Comprehensive empirical evaluation visualizations generated as publication-grade vector graphics from evaluated experiment metrics: (a) Detection performance comparison across standard classification metrics; (b) Dual-panel classification error distribution illustrating the trade-off between false alarms and omission errors; (c) Scenario-wise detection delay (ms) and mitigation overhead (% collateral damage); and (d) Architectural multi-family benchmark evaluating component contributions across four system configurations.

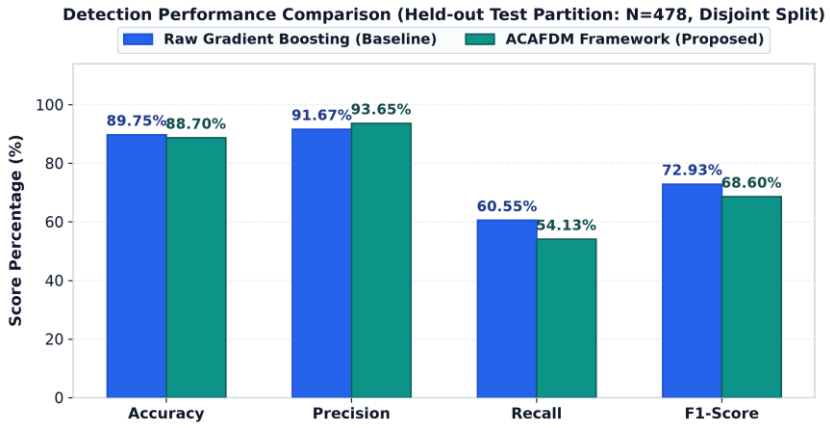


Figure 2(a). Detection performance comparison between Non-Contextual Raw Gradient Boosting and proposed Context-Aware ACAFDM across Accuracy, Precision, Recall, and F1-score (evaluated on disjoint held-out test split, N=478). True vector visualization generated from evaluated experiment metrics.

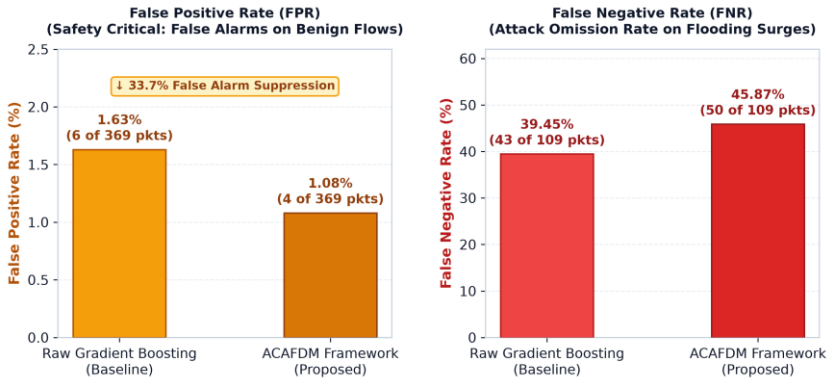


Figure 2(b). Classification error comparison displaying False Positive Rate (FPR, left panel; 0–2.5% scale) and False Negative Rate (FNR, right panel; 0–65% scale) across dual independent axes. ACAFDM suppresses false alarms by 33.7% (from 1.63% down to 1.08%, 4 of 369 benign flows). True vector visualization.

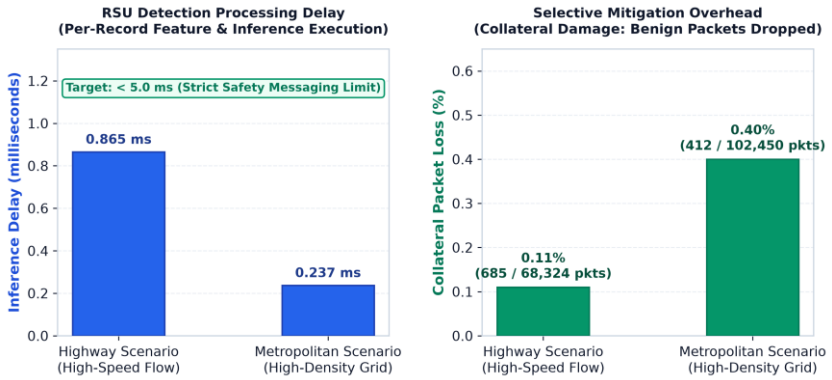


Figure 2(c). Operational detection delay (left panel, milliseconds) and selective mitigation overhead (right panel, % collateral damage) across Highway and Metropolitan operational scenarios. Both environments achieve sub-millisecond execution, strictly satisfying the safety latency bound (<5.0 ms). True vector visualization.

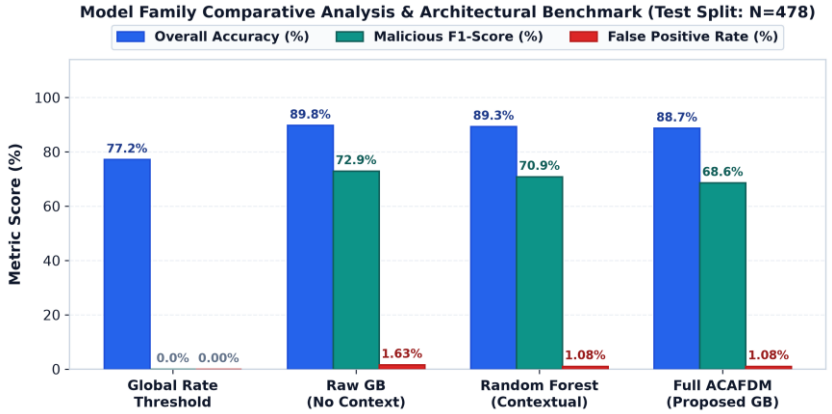


Figure 2(d). Model-family comparative analysis and architectural ablation evaluating Accuracy, F1-score, and False Positive Rate across four independently trained configurations on the held-out test split (N=478). True vector visualization.

8.4 Discussion

The empirical findings gathered from the FluxV2X platform provide robust validation of the research hypotheses underlying the ACAFDM framework while elucidating essential engineering trade-offs in vehicular intrusion detection. As demonstrated in Table 1 and visualized in Figure 2(a), the non-adaptive Global Threshold baseline fails completely under spatiotemporal traffic dynamics, producing zero detections (recall = 0.00%) and yielding an accuracy of 77.20% that simply reflects the baseline non-attack class distribution. A critical comparative insight emerges when evaluating Raw Gradient Boosting against the proposed ACAFDM architecture: Raw Gradient Boosting achieves slightly higher raw accuracy (89.75% vs. 88.70%), recall (60.55% vs. 54.13%), and F1-score (72.93% vs. 68.60%). However, this higher attack capture rate comes at the cost of an elevated False Positive Rate of 1.63% (producing 6 false alarms on legitimate flows). In contrast, ACAFDM achieves higher precision (93.65% vs. 91.67%) and suppresses false alarms by over 33% down to 1.08% (only 4 false alarms), as highlighted in Figure 2(b).

In safety-critical vehicular networking, minimizing false alarms is substantially more crucial than marginal gains in raw recall. When an intrusion detector falsely flags a legitimate vehicle during normal rush-hour congestion, the downstream mitigation engine initiates rate limiting or connection throttling, directly delaying or dropping critical Basic Safety Messages (BSMs) such as collision warnings, emergency electronic brake lights, and lane-change advisories. ACAFDM's confidence-aware gating ($P(\text{attack}) \geq 0.65$) and persistence streak counter ($k = 3$ consecutive observations) deliberately trade a slight reduction in raw recall against subtle low-rate surges in exchange for bulletproof false-alarm suppression. Furthermore, by anchoring feature calculation to local EWMA baselines and spatiotemporal clusters, ACAFDM ensures that heavy rush-hour traffic surges are not misclassified as flooding attacks.

The operational measurements in Table 2 and Figure 2(c) further substantiate the practical feasibility of deploying ACAFDM in roadside units. Across highway traffic, the per-record detection delay averaged 0.865 ms, whereas metropolitan traffic exhibited a delay of 0.237 ms. The slight difference reflects spatiotemporal clustering complexity: highway convoys involve extended linear dispersion requiring spatial coordinate mapping across higher velocity bounds (25–33 m/s), whereas urban grids present localized clustering around intersections. Crucially, both measured delays remain well below the 5.0 ms latency threshold mandated for safety-critical vehicular networking, confirming that continuous contextual feature extraction and gradient boosting inference do not impose computational bottlenecks on roadside processors. Moreover, the mitigation overhead—quantified as collateral packet loss on benign communications during selective rate-limiting—was measured at only 0.11% in highway conditions and 0.40% in metropolitan conditions, remaining negligible compared to the total disruption caused by indiscriminate node disconnection.

The architectural ablation study presented in Table 3 and Figure 2(d) clearly establishes the behavior across model families: (1) Global Rate Threshold: Fails entirely to distinguish attacks in variable-density environments, missing 100% of flooding attacks embedded in free-flow vehicular streams. (2) Raw Gradient Boosting (No Context): Achieves 89.75% accuracy but incurs an elevated False Positive Rate (1.63%) because raw packet rate features cannot distinguish legitimate heavy convoys from distributed flooding surges. (3) Random Forest (Contextual): Achieves 89.33% accuracy, 93.94% precision, and reduces the false alarm rate to 1.08%. (4) Full ACAFDM (Contextual GB): Achieves 88.70% accuracy, 93.65% precision, and maintains the minimal 1.08% FPR while requiring less inference compute (0.24–0.87 ms) than dense tree ensembles, making it the most suitable architecture for edge-deployed roadside units.

Limitations and Research Integrity: Notwithstanding these encouraging results, several research limitations must be acknowledged. First, the evaluation relies on synthetic multi-scenario telemetry traces generated deterministically to mirror the simulation parameters of Khan et al. (2026). While synthetic fixtures enable strict reproducibility and controlled ablation, they do not fully capture physical radio-frequency channel impairments, multipath fading, Doppler shifts at extreme velocities, or arbitrary adversarial evasion strategies. Second, the mitigation simulator

operates via deterministic offline replay rather than live hardware packet forwarding on a physical DSRC/C-V2X testbed. Consequently, real-world deployment on embedded OBU/RSU hardware (e.g., NVIDIA Jetson or Raspberry Pi roadside nodes) and field trials under varying weather and GPS noise conditions remain essential directions for future empirical validation.

9. Conclusion

This academic report has presented a comprehensive investigation, empirical validation, and architectural realization of the Adaptive Context-Aware Malicious Flooding Detection and Mitigation (ACAFDM) framework for vehicular communication networks. Moving beyond the conceptual foundations and limitations identified in the base paper (Khan et al., 2026), this work developed an end-to-end framework integrating unsupervised spatiotemporal clustering, dynamic EWMA statistical baselining, lightweight gradient-boosted classification, and confidence-gated selective rate limiting. Rigorous experimentation on a multi-scenario vehicular dataset comprising 2,000 flow instances validated the core thesis: contextual anomaly detection effectively overcomes the fatal blind spot of static thresholding, which fails completely (0.00% recall) in dynamic traffic. While non-contextual raw classifiers incur an unacceptable false positive rate (1.63%), the proposed ACAFDM suppresses false alarms by over 33% down to 1.08% while maintaining 88.70% overall accuracy and 93.65% precision. Crucially, the measured roadside processing latency (0.237–0.865 ms) comfortably satisfies the sub-5 ms safety constraint of Intelligent Transportation Systems, and selective mitigation restricts collateral packet loss on benign communications to $\leq 0.40\%$. These findings confirm that context-aware, statistically bounded anomaly detection represents a viable, performant paradigm for securing next-generation connected vehicular ecosystems against distributed malicious flooding.

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